APM1205 - Linear Models
Summative Assessment 1

Introduction and research problem

The Seoul Bike Dataset was imported into Posit Cloud R and examined to understand its structure and variables. The categorical variables, namely Season, Holiday, and Functioning Day, were converted into factors. The main objective is to determine how weather conditions and seasonal changes affect the number of bike rentals.

Variables used in analysis:

- Rented Bike Count →Rentals
- Temperature (°C)
- Humidity (%)
- Wind speed
- Seasons

Dataset description

The dataset contains information on daily bike rental activity along with weather conditions in Seoul. The variables selected include the number of rentals, temperature, humidity, wind speed, and its season. These variables are expected to influence an individual's willingness to use bike rental services.

Research questions/hypotheses

- **Null Hypothesis (H_0):** There is no relationship between the temperature and the number of bike rentals.
- **Alternative Hypothesis (H_1):** There is a significant relationship between the temperature and the number of bike rentals.

- **Null Hypothesis (H_0):** Humidity, wind speed, and season do not affect bike rentals.
- **Alternative Hypothesis (H_1):** At least of these variables significantly affects bike rentals.

- **Null Hypothesis (H_0):** There is no interaction between temperature and season.
- **Alternative Hypothesis (H_1):** There is a significant interaction between temperature and season.

Exploratory data analysis

Descriptive statistics were computed for the variables Bike Rentals, Temperature, Humidity, and Wind Speed. It measures the mean, median, and the standard deviation of bike rentals.

A histogram was created to examine the distribution of rental demand. It shows the rental counts varied across several observations, indicating that demand changes according to weather and seasonal conditions. Scatter plot shows if there is any relationship between the bike rentals and the weather variables. The Temperature versus Rentals plot shows that there is a positive relationship, where higher temperatures increased bike rental demand. On the other hand, the Humidity versus Rentals plot suggests that higher humidity levels may reduce bike usage. As shown in the boxplot result, plot reveals noticeable differences among seasons, indicating that rental demand is not constant throughout the year. Overall, this analysis provided evidence that weather conditions and seasonal factors influence bike rental demand in Seoul.

Simple linear regression

The regression results showed that as temperature increases the bike rentals tend to increase as well. The line displayed an upward, supporting this positive relationship. This suggests that temperature is an important predictor of bike rental demand. Therefore, it concludes that warmer weather in Seoul tends to increase bicycle usage.

Multiple linear regression

The multiple linear regression incorporated some variables including temperature, humidity, wind speed, and season as explanatory variables. The inclusion of these variables improved the model's ability to explain variations in bike rental demand. The result showed that bike rental is affected by a combination of these factors rather than temperature alone.

Dummy-variable regression

At least one season was treated as the reference category, while the remaining seasons were represented by dummy variables. This approach allowed the seasonal effect on bike rental demand to be interpreted within the regression.

Interaction analysis

To determine whether the effect of temperature differs across seasons , an interaction model was created. This model included both temperature and season. Values were calculated and plotted for each season, resulting in separate lines representing each season. The differences in lines suggests that the effect of temperature on bike rentals may vary depending on the season, indicating a possible interaction between seasonal conditions and temperature.

Conclusion

This study analyzes the factors affecting bike rental demand using the Seoul Bike dataset. This revealed that variables including temperature, humidity, wind speed, and season are associated with bicycle rental activity. It clearly showed that there is a positive relationship between temperature and bike rentals using simple linear regression. Also, multiple linear regression also used additional variables to see how the seasonal differences affect bike rental behavior. The analysis suggested that the impact of temperature may vary across seasons. Overall, temperature was identified as one of the most influential predictors, while the other variables also contributed to variations of rental activity.